5 **(a)** A magnet produces a uniform magnetic field of flux density *B* in the space between its poles.

A rigid copper wire carrying a current is balanced on a pivot. Part PQLM of the wire is between the poles of the magnet, as illustrated in Figure 5.1.

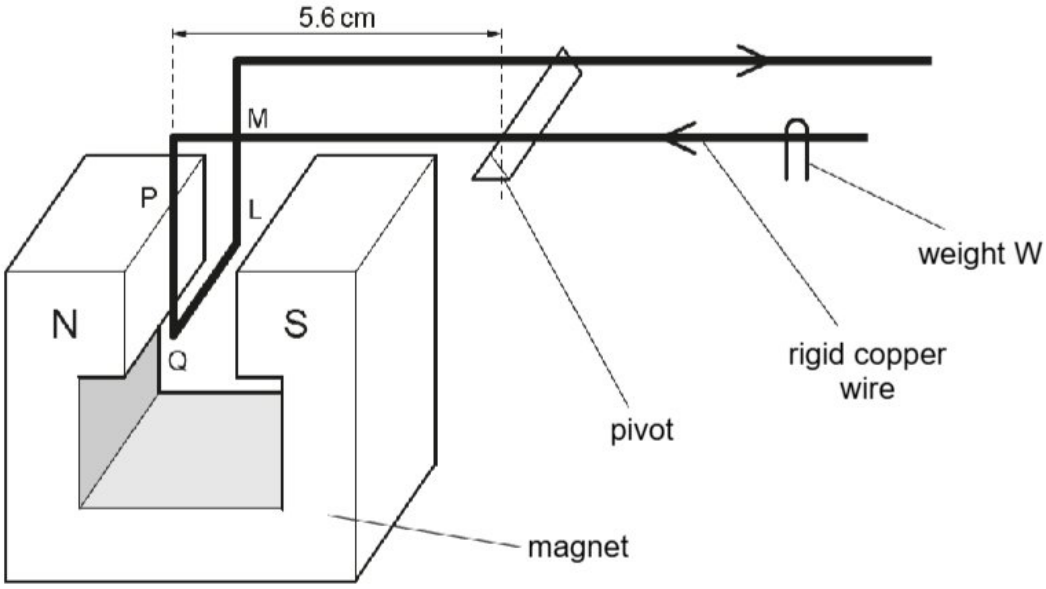


Figure 5.1 (not to scale)

The wire is balanced horizontally by a small weight W whose mass is 1.3×10^{-4} kg.

Section QL of the wire has length 0.85 cm.

The perpendicular distance of QL from the pivot is 5.6 cm.

When the current in the wire is changed by 1.2 A, W is moved a distance of 2.6 cm along the wire in order to restore equilibrium.

(i) Show that the change in moment of W about the pivot is 3.3×10^{-5} N m.

[1]

(ii) Use the information in **5a(i)** to determine the magnetic flux density *B* between the poles of the magnet.

B = ...T [2]

(b) In a different experiment, electrons are accelerated through a potential difference and then enter a region of uniform magnetic field of flux density 1.5 mT.

Figure 5.2 illustrated such an electron travelling at right angles to the uniform magnetic field.

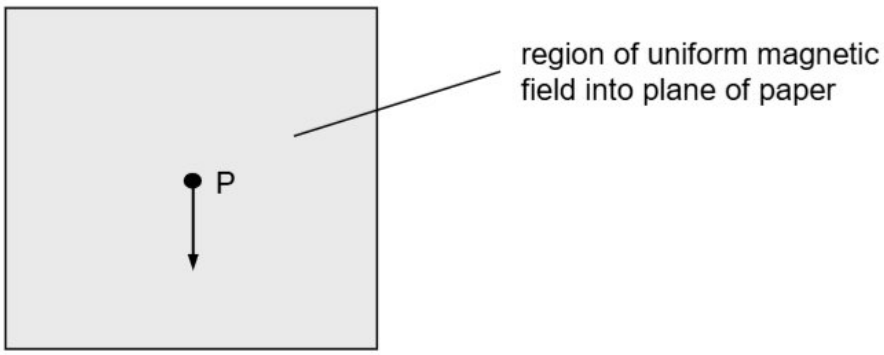


Figure 5.2

The magnetic field is directed into the plane of the paper.

When the electron is at P, its velocity is 2.9×10^7 m s^{-1} in the direction shown. This is normal to the magnetic field.

(i) On Figure 5.2, sketch the path of the electron, assuming that it does not leave the region of the magnetic field.

[2]

(ii) Calculate the radius of the path of the electron. Explain your working.

radius = ...m [2]

(iii) A uniform electric field is now produced in the same region and in the same direction as the magnetic field. Suggest the shape of the resultant path of the electron. You may draw a sketch to illustrate the path, if you wish.

.....

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..... [2]

[Total: 9]